

RESEARCH ARTICLE

Conditional cash transfers, spillovers, and informal health care: Evidence from Peru

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Salaverry 2020, Lima 11, Peru.
Email: winkelried_dm@up.edu.pe**Abstract**

Conditional cash transfer programs have become instrumental in encouraging the use of formal health services in developing countries, but little is known about their effect on the use of low-quality informal care. Using a large survey of Peruvian rural households and a regression discontinuity design, we find a sizeable reduction in the use of informal health care providers not only by targeted but also by nontargeted members of households that qualify for the program. This indicates the existence of spillover effects within the household. We also provide evidence that beyond the direct increase in income, the availability of better information about institutional services is a potential mechanism that drives these effects.

KEYWORDS

cash transfer program, informal health care providers, regression discontinuity

JEL CLASSIFICATION

C21; I18; O17

1 | INTRODUCTION

Informal providers account for a substantial proportion of the health care in developing countries (cf. Cross & MacGregor, 2010; Bloom et al., 2011). Governmental efforts to promote the use of institutional services instead of lower quality care obtained from informal providers have not always been effective (cf. Dupas, 2011; Randive, Diwan, & De Costa, 2013). Affordability, cultural biases, a mistrust in modern medicine, and the lack of information about the low-quality services are the main factors behind the demand for care from informal providers (cf. Sudhinaraset, Ingram, Lofthouse, & Montagu, 2013).¹

Demand-side programs, such as conditional cash transfers (CCTs), might become a key instrument to address this issue. A large body of literature gives substantial evidence on the positive effects of CCT programs on several health outcomes (see, inter alia, Gertler, 2004; Morris, Flores, Olinto, & Medina, 2004; Behrman & Hoddinott, 2005; Lagarde, Haines, & Palmer, 2007; Barham & Maluccio, 2009; Evans, Holtemeyer, & Kosec, 2017; Parker & Todd, 2017).² However, little is known about the impact of CCT programs on the use of care from informal providers.³ Moreover, the literature

¹Besides encouraging the use of formal health services, a complementary policy to improve the supply of affordable health care in developing countries might be to increase the quality of informal care through formal training (see, inter alia, Das, Chowdhury, Reshmaan, & Banerjee, 2016).

²Most studies focus on the use of services that are part of the program's health requirements. Few papers analyze if CCT programs also increase the use of preventive health care services (cf. Witvorapong & Foshanji, 2016) or the likelihood of seeking institutional treatment when the children fall ill (cf. Evans et al., 2017).

³It is not necessarily obvious that the evidence that CCT programs increase the use of formal health services represents a net substitution of informal providers or, given the increase in income due to the CCT, a greater demand for both types of services (i.e., formal and informal). For instance,

often focuses only on the effects on target populations (i.e., children and pregnant women), neglecting the potential for behavioral responses from other family members. The evidence about spillover effects of CCT programs on health outcomes is scarce (see, *inter alia*, VerPloeg, 2009; Shei, Costa, Reis, & Ko, 2014; Contreras & Maitra, 2013). This evidence is particularly important because whether these programs are able to expand their positive effects to nontarget members of treated households is crucial for the programs to be effective in encouraging the use of institutional health services.

This paper studies the effect of a CCT program on the demand for informal health care once users fall ill. We analyze not only the effect on the target population but also the spillover effect on nontarget members of the household. To the best of our knowledge, this is the first study to analyze the direct relation between CCT programs and the use of informal health care services and to explore the spillover effects. Our main contributions are to show that CCTs can be effective in reducing the use of informal health care providers, regardless of whether individuals are the target population of the program or not, and to point out an income effect and the availability of better information on formal health care as potential mechanisms behind such a reduction. Understanding these effects can provide new insights for policymakers on how CCT programs can help reshape health systems in poor countries.

We study the Peruvian CCT program, named *Juntos*. Since 2012, the government has based eligibility for the program on the value of a socioeconomic score. We can compute this score from the information in household surveys and then verify that when it crosses a threshold, it generates a discontinuity in the probability of participation in the program. Thus, we can estimate the causal effects of the program using a “fuzzy” regression discontinuity design (see Lee & Lemieux, 2010).

Using a large survey of rural households, we find robust evidence that the CCT program reduces the use of informal health care providers. In particular, the probability that an ill individual from the target group will seek medical attention at such a provider is reduced by 0.25 to 0.37. This finding indicates that families with CCT beneficiaries develop the habit of attending the formal facilities, as they become more familiar with it and its staff, which consequentially reduces the usage of informal providers. Further, we observe a somewhat smaller, but still significant, effect (between 0.22 and 0.29) on nontarget members. The very habit of attending a formal facility extends to all family members, not only the program beneficiaries.

We also provide evidence that beyond the direct increase in income, the availability of better information about institutional services and the greater confidence generated by the frequent interaction with formal health facilities are potential drivers of the program effects. Effects are stronger if the ill individual belongs to (a) a household without land (i.e., poorer households in rural areas), (b) a beneficiary household with young children of less than 7 years of age (i.e., program beneficiaries with frequent checkups), or (c) a household that is targeted by another social program (i.e., more used to social services). In all these cases, the income and informative shocks that are generated by participation in the CCT program play an important role in our main results.

This paper complements the literature on the effect of CCT programs on the use of incentivized, formal health services (see Gertler, 2004; Lagarde et al., 2007; Attanasio, Oppedisano, & Vera-Hernandez, 2015; Parker & Todd, 2017; Díaz & Saldarriaga, 2019) by estimating their effect on the use of nonincentivized, informal health services. Second, it contributes to a growing literature on the spillover effects of social programs on health outcomes (see Handa, Huerta, Perez, & Straffon, 2001; Miguel & Kremer, 2004; Banerjee, Duflo, Glennerster, & Kothari, 2010; Avitabile, 2012; Baird, De Hoop, & Özler, 2013; Benjamin-Chung et al., 2017) by exploring the effects of the CCT program on members of the treated household that do not belong to the target population.

The rest of the paper is organized as follows. Section 2 provides some background on the Peruvian CCT program and presents its assignment rule that is key to the identification of the treatment effect of interest. Section 3 describes the econometric framework and presents the data. Section 4 shows the main empirical findings for the direct effects on the target population and for the spillover effects on the nontarget population, along with a number of robustness checks. This section also explores the heterogeneity in the effects across several population groups to investigate the potential mechanisms that underlie our results. Section 5 concludes and suggests some avenues for further research. A number of additional robustness checks and further empirical exploration are included in the Supporting Information.

2 | INSTITUTIONAL BACKGROUND

Driven by the successful experience of the Mexican program Progres/Oportunidades (see, *inter alia*, Parker & Todd, 2017), Peru launched its own CCT program, called *Juntos*, in 2005. The program has operated ever since and is considered

the most important social program in the country, in terms of both coverage (it benefits around 70% of rural households) and budget (about 22% of all social programs budget). Since 2012, the program is managed by the Ministry of Social Inclusion and Development (MIDIS).

A growing body of literature has studied the effects of this CCT program on various well-being indicators. Perova and Vakis (2009) present early evidence on positive but small effects on the expenditure of nutritious food and on school enrollment. Gahlaut (2011) finds significantly negative effects on the incidence of child labor. Sánchez, Meléndez, and Behrman (2018) find that the program reduces severe stunting and improves height-for-age ratios among treated children between the ages of 5 and 7. Perova and Vakis (2012) find the positive effects of the program to be stronger for longer treatment spells; yet some effects remain small in magnitude, which indicates that the program has room for improvement. Pérez-Lu, Cárcamo, Nandi, and Kaufman (2017) show that *Juntos* reduces the number of underweight pregnant women and also reduces anemia among preschoolers. Finally, Díaz and Saldarriaga (2019) find that the program increases prenatal care utilization among eligible women, thereby increasing the probability of receiving prenatal care delivered by a professional and, in turn, reducing the probability of obstetric emergencies during labor and childbirth.

2.1 | Conditionalities and benefits

The CCT program's objective is to promote and speed up the access to quality social services by acknowledging not only poverty but also other sources of vulnerability such as political violence and disabilities. The target population are children or teenagers up to 19 years old, pregnant women, breastfeeding mothers, and people with disabilities. The program's transfers are conditional on the targeted household's members meeting a number of requirements, namely, school enrollment and attendance, and attending regular nutrition and health checkups. It is worth mentioning that the conditionalities on health care use are very limited: *Juntos* only requires pregnant women and children between 0 and 3 years of age to attend regular medical checkups in the Ministry of Health (MINSA) facilities (see Programa Juntos, 2016, for further details).⁴

The recipient of the cash transfer is usually the children's mother, the woman in charge of the children, or the beneficiary woman. Preference is given to female recipients due to the abundant evidence that they are most likely to devote the extra income toward the well-being of the family (see, inter alia, Bergolo & Galván, 2018). The government sends the money transfer monthly or every 2 months. Households receive S/ 100 (about US\$30) every month or S/ 200 (about US\$60) every 2 months. These amounts are relevant for rural families: They represented 44% of the rural poverty line (which is a per capita figure) in 2014 or, according to Sánchez and Rodríguez (2016), about 13% of the average household's expenditures.

2.2 | Eligibility

Eligibility to *Juntos* is based on a number of criteria in a three-stage procedure (see Perova & Vakis, 2012). The first stage is geographic targeting: all households in districts (municipalities) with a poverty rate above 40% pass to the next stage. The National Statistical Office determines the poverty rate by using a Poverty Map *à la* Elbers, Lanjouw, and Lanjouw (2003) that is computed with a combination of data from the Census and the National Household Survey (ENAHU).

Then, the MIDIS computes a socioeconomic score, the “household targeting index” (IFH), as a weighted average of a large set of preestablished predictors of poverty.⁵ For rural areas, the main variables include the fuel used for cooking, the number of members with health insurance, asset ownership, the education level of the household head, the highest level of education among any member of the household, access to electricity, and the flooring material. To determine eligibility, the country is divided into 15 clusters defined by areas with similar monetary poverty, and the IFH is compared with cluster-specific thresholds. These thresholds can be further deducted from the IFH to normalize the cutoff point to zero. The final score ranges from −4 to 4.⁶

⁴Children between 0 and 3 years old and pregnant women are not required to attend formal health facilities when they are ill. Older children and adolescents, which may also be beneficiaries, are not required to attend medical checkups and are not required to use formal facilities in the case of illness either. There are no requirements whatsoever for the rest of the household members.

⁵The IFH is used for the assignment of various social programs in Peru. Bernal, Carpio, and Klein (2017) study the case of social health insurance and provide a very detailed and comprehensive online appendix on the computation of the IFH (appendix F).

⁶Since 2012, following its creation, the MIDIS made public all the necessary information for third parties (i.e., researchers) to infer the eligibility status of individuals from the information in household surveys (see Escobal & Benites, 2012).

The second stage is household targeting, based on three criteria: (a) The IFH classifies the household as poor (i.e., $IFH > 0$); (b) at least one member of the household belongs to the target population; and (c) the household is considered poor according to the administrative (and confidential) data on income. Eligible families have to demonstrate that they had lived in the same district for the last 6 months. Also, they have to register with the health center and the school where the conditions are to be fulfilled.

The last stage is community validation. The MIDIS staff meet with a number of community members, local authorities, and representatives of other ministries (notably, education and health) to validate the results of the second stage and to update the administrative records. The purpose is to reduce inclusion and exclusion errors, because at this stage, a household can prove, through updated income records, that it needs the transfer despite $IFH < 0$; analogously, a household can be considered ineligible despite $IFH > 0$. There is evidence that the program has some problems with leakage as well as undercoverage, though not as severe as other Peruvian social programs (see Programa Juntos, 2016).

Further, there are some exceptions where only geographic targeting applies to politically sensitive areas. In particular, all households that are affected by violence from drug trafficking and terrorism (such as the region named VRAEM) or districts that protect indigenous communities are eligible regardless of the values of the IFH. Also, households with members with disabilities are automatically eligible.

3 | METHODS AND DATA

3.1 | Methodology

We are able to compute the IFH from the information in household surveys (see, for instance, Bernal et al., 2017) and to sort households accordingly. Also, we observe whether a household is a beneficiary of the CCT program or not. However, although the IFH predicts whether a household is a *Juntos* beneficiary, it does so imperfectly. Thus, we use a “fuzzy” regression discontinuity design (RDD).

With observational data, individuals self-select into the treatment. However, in an RDD, a given rule depends on the value of a running variable relative to a cutoff point to assign the treatment. Near the cutoff point, the assignment can be considered “as good as random,” that is, treated and control units are comparable in all aspects but the treatment. But, as shown in Hahn, Todd, and Van der Klaauw (2001), in a “fuzzy” RDD, the running variable cannot predict the treatment perfectly. However, the eligibility rule still generates a discontinuity at the cutoff point in the probability of treatment, so the treatment effect can be identified using eligibility as an instrument for the treatment.

Let Y be the outcome of interest (i.e., the use of an informal health care provider when falling ill), D be the CCT treatment dummy variable ($D = 1$ when treated, $D = 0$ otherwise), X be the running variable (i.e., the IFH) whose cutoff point is set with no loss of generality to zero, and T be the eligibility dummy variable ($T = 1$ when $X \geq 0$, $T = 0$ when $X < 0$). The model for the outcome and treatment is as follows:

$$\begin{aligned} Y &= \alpha_1 + \tau D + f(X) + \varepsilon_1, \\ D &= \alpha_2 + \delta T + g(X) + \varepsilon_2, \end{aligned} \quad (1)$$

where $f(\cdot)$ and $g(\cdot)$ are flexible functions of X , ε_1 and ε_2 are two possibly correlated error terms, and τ is the treatment effect of interest. Hahn et al. (2001) show that if $f(\cdot)$ and $g(\cdot)$ are of the same form (specifically, the same polynomial degree and bandwidth), then τ and its standard error can be consistently estimated from an instrumental variable regression of Y on D using T as an instrument, controlling for flexible functions of X to remove the effect of other factors that vary smoothly around the cutoff point.

Following, inter alia, Gelman and Imbens (2019), the function $f(\cdot)$ is parameterized as follows:

$$f(X) = \beta_1 X + \beta_2 X^2 + \dots + \beta_d X^d + T \times (\gamma_1 X + \gamma_2 X^2 + \dots + \gamma_d X^d) \quad (2)$$

for some polynomial degree d . The inclusion of interaction terms between T and the powers of X allows the regression function to differ on both sides of the cutoff point. Function $g(\cdot)$ is specified analogously.

Because the treatment effect is local (i.e., around the discontinuity), it is often advisable to constrain the sample to observations within narrow bandwidths of the cutoff point (see, inter alia, Imbens & Kalyanaraman, 2012). In particular, for bandwidths h_L and h_R , the sample is such that $h_L \leq X \leq h_R$; to simplify matters, we use $-h_L = h_R = h$.

The choice of d and h involves a trade-off between bias and efficiency where higher order polynomials and smaller bandwidths reduce the bias but at the cost of greater asymptotic variance (see Lee & Lemieux, 2010). The choice of d is a standard model selection issue assisted by information criteria. Gelman and Imbens (2019) show that, due to overfitting, estimates might become very sensitive and misleading when d is large. Therefore, as a rule of thumb, they advise using $d \leq 2$. On the other hand, the choice of h is based on the minimization mean-square criteria, as advanced in Calonico, Cattaneo, and Titiunik (2014).

3.2 | Data

We use two main data sources. First, we use the National Registry of Social Programs Users that provides the MIDIS's administrative records on the beneficiaries and the enrollment dates for all Peruvian social programs.

Second, we use ENAHOR 2014, which is a special edition of the household survey ENAHO that is dedicated to rural areas. Both surveys are managed by the National Statistical Office, are nationally representative, and are aimed at measuring the poverty rate and general living conditions. They contain detailed information on education, employment, income and expenditure, housing, and perceptions that are required to compute the IFH and to control for socioeconomic characteristics. The most important difference between surveys is that ENAHOR 2014 is representative at the rural provincial level, whereas ENAHO is representative at the department (i.e., a collection of provinces) level only. The reason is sheer sample size as ENAHOR 2014 is about 10 times as large as the typical ENAHO: respectively, 110,181 households versus 9,770 rural households in total and 55,407 versus 6,691 households in target *Juntos* rural districts.

Regarding health outcomes, the surveys first asks individuals what their health status was during the last month (the head of the household or his or her partner answers in the case of children). In particular, we consider individuals as ill if they experienced a severe illness (such as influenza or colitis), the relapse of a chronic disease, or an accident. Minor discomforts such as headaches, fever, or nausea are not considered an illness.⁷

In addition, ill individuals also indicate if they sought medical attention and where or with whom. Patients can choose a formal provider from a long list of public health care facilities and centers that are administered mainly but not exclusively by the MINSA. Or they can choose private emergency centers, clinics, polyclinics, or the doctor's office. Moreover, they can also choose from informal providers that include an apothecary or a pharmacy, unidentified relative or friend, unlicensed chiropractor (*huesero*), traditional healer (*yerbero* or *curandero*), or other unspecified providers.

To construct our final database, we match the data from ENAHOR 2014 to the administrative records to identify the CCT program beneficiaries. This amounts to 319,930 individuals in 110,181 households. Then, we restrict the sample following the eligibility criteria outlined in Section 2.2: First, we consider only target districts (209,620 individuals); second, we exclude VRAEM areas due to their exceptionality (rendering 203,645 individuals); third, we keep households with at least one individual from the target population, that is, children, adolescents, or pregnant women (168,044 individuals); fourth, because we can determine eligibility only for individuals that became *Juntos* recipients since 2012, we only include households with an enrolment date between 2012 and 2014, the year of the survey (94,185 individuals); and finally, we exclude members older than 65 years of age, because poor elderly are eligible individually for another transfer program (*Pensión 65*). As a result, the sample has 89,585 individuals (in 18,073 households), out of which 15,357 reported being ill during the last month: 5,732 from the target population and 9,625 from the nontarget population.

Both equations in (1) can be augmented with control variables without altering in any significant way the methodological framework used for inference. As control variables, we use individual characteristics such as a female dummy variable, age and age squared, maximum education level achieved, an illiterate dummy variable, and a set of dummy variables for the mother tongue (Spanish, Quechua, Aymara, and others). Also, we add a dummy variable for whether the district is considered an indigenous community. Importantly, we also consider three supply-side variables akin to the public health provision in the country: first, the reported time it takes to get from the house to the closest formal health center administered by MINSA; second, the Euclidean distance, that we obtain from the geographical coordinates, between the closest health center (the georeferencing is provided by MINSA) and the house (the coordinates are given in ENAHOR 2014); and third, an indicator of whether the household is covered by public health insurance. Some descriptive statistics are given in Table 1.

⁷In Section B.2 of the Supporting Information, we show that our main results remain robust to alternative definitions of illness.

TABLE 1 Descriptive statistics

Variable	Eligible				Noneligible				All	
	Target		Nontarget		Target		Nontarget		All	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
<i>Juntos</i> (first stage)	0.943	0.231	0.950	0.217	0.498	0.500	0.425	0.494	0.714	0.452
(a) Outcomes										
Attendance	0.781	0.414	0.794	0.403	0.802	0.399	0.828	0.377	0.803	0.398
Informal care	0.241	0.428	0.248	0.432	0.230	0.421	0.242	0.429	0.242	0.428
Self-reported health status	0.169	0.375	0.073	0.260	0.196	0.397	0.112	0.316	0.125	0.331
(b) Individual characteristics										
Female	0.519	0.499	0.564	0.496	0.499	0.500	0.534	0.499	0.534	0.498
Age	10.4	5.2	41.9	12.0	10.1	5.3	40.6	11.6	29.7	18.0
Preschool education	0.164	0.371	0.152	0.359	0.149	0.356	0.057	0.232	0.126	0.331
Primary education	0.491	0.500	0.365	0.481	0.508	0.500	0.228	0.419	0.375	0.484
Secondary education	0.344	0.475	0.483	0.499	0.343	0.475	0.715	0.452	0.499	0.500
Literacy	0.373	0.484	0.406	0.491	0.238	0.426	0.285	0.451	0.334	0.472
Mother tongue is Spanish	0.689	0.463	0.477	0.499	0.906	0.291	0.787	0.410	0.686	0.464
Indigenous community	0.211	0.408	0.229	0.421	0.161	0.368	0.125	0.330	0.183	0.387
(c) Health supply indicators										
Distance to closest health center (km)	7.834	0.527	7.769	0.516	8.079	0.401	8.036	0.393	7.915	0.484
Distance to closest health center (min)	89.406	81.025	89.183	80.294	77.599	72.829	76.619	69.751	83.412	76.361
Public health insurance	0.855	0.362	0.780	0.414	0.741	0.438	0.638	0.480	0.745	0.435
Observations	2,987	2,987	5,105	5,105	2,745	2,745	4,520	4,520	15,357	15,357

Source: ENAHOR 2014. Own elaboration. *Note.* These statistics are for the sample of individuals that reported illness (such as influenza or colitis, the relapse of a chronic disease, or an accident).

4 | RESULTS

4.1 | Baseline results

In our empirical analysis, we examine whether the CCT program affects health care outcomes that are not in the set of the *Juntos* conditionalities (e.g., an ill 5 years old is not required to be treated in a MINSA facility).⁸ Health care use is reported only by those who report illness. We first test and fail to reject the null hypothesis that the program has no effect on reported illness (see Appendix A). Hence, there is no risk of selection bias in limiting attention to the sample that reports illness, and we proceed to do this.

In this sample, we do not find any significant effect of the program on the use of any health care without distinguishing between formal and informal care (see Appendix B). However, we do find significant results when it comes to choosing between formal and informal care. Given the finding of no significant effect on the use of any health care, a decrease in the usage of informal care comes in tandem with an increase in the usage of formal care of about the same magnitude (see Appendix A3). Therefore, we only present the results on the use of informal health care.

We estimate the program's effects on the use of care from informal providers for two subsamples. The results for the target population quantify the direct effect of the program, whereas the estimations for the nontarget population measure a spillover effect within members of the household. It is worth mentioning that by construction of our sample, all households have members from the target and the nontarget population. Thus, the spillover effects come from the comparison between nontarget members from a nontreated (but eligible) household with nontarget members from a treated household.⁹

Figure 1 plots the regression discontinuities for the first-stage equation (i.e., D as a function of X) and the reduced forms for the probability of using an informal care provider when ill (i.e., Y as a function of X) for both the target and the

⁸In Section B.3 of the Supporting Information, we show the empirical results we report below to be similar to the results on a sample that excludes the households with members that must comply with the health-related conditionalities of the program. Therefore, the effects we identify are well beyond a mere manifestation of the mechanical compliance of the *Juntos* health conditionalities.

⁹This implementation of the spillover effect within the household has been previously used in Qureshi (2016) for educational outcomes and Adams et al. (2018) for health outcomes.

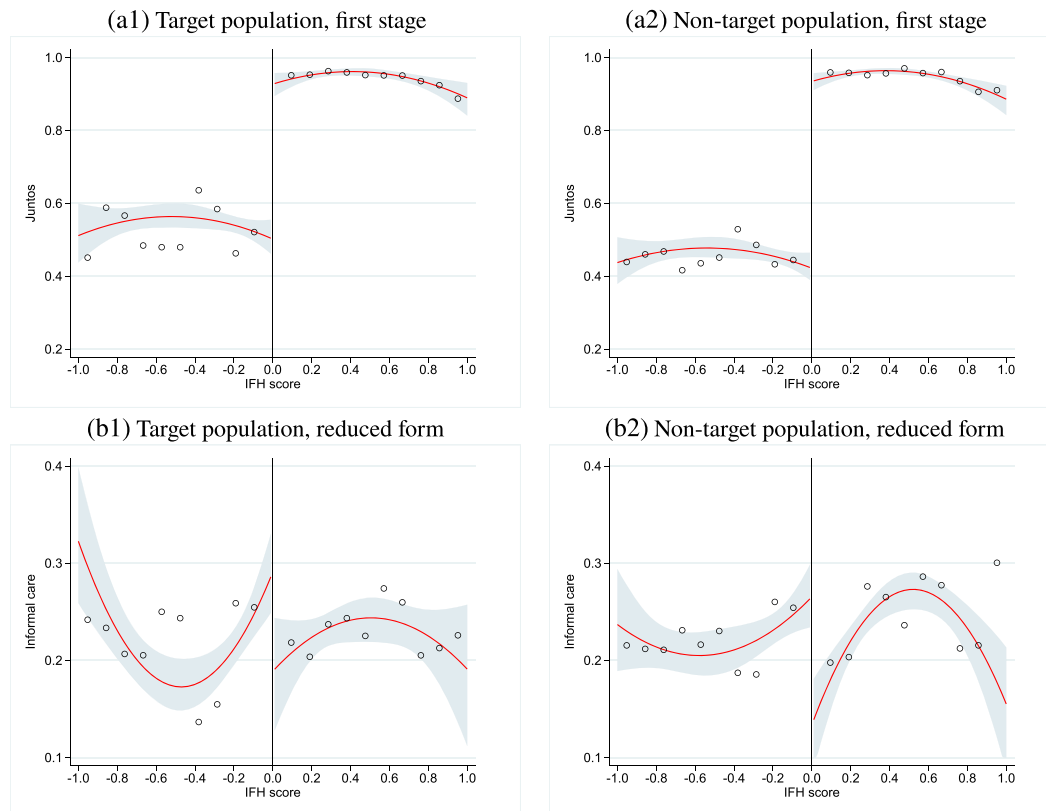


FIGURE 1 Regression functions for the first stage and reduced forms. The estimates of the regression functions of the first stage (i.e., a regression of D on T and a flexible function of X) are in panels a1 and a2, and the reduced form (i.e., a regression of Y on T and a flexible function of X) are in panels b1 and b2. These use for $d = 2$ and $h = 1$, which is approximately optimal in the sense of Calonico et al. (2014), and results for alternative values of d and h are similar. In all panels, the horizontal axis is the IFH score; in panels a1 and a2, the vertical axis is the proportion of households treated by the *Juntos* program, and in panels b1 and b2, the vertical axis is the proportion of individuals using an informal provider. The shaded areas are 95% confidence intervals. The circles are averages of D or Y across intervals of X , each of width $1/20$. Source: ENAHOR 2014. Own elaboration [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/hec.3956)]

nontarget populations.¹⁰ In both cases, the first stage reveals a discrete, substantial jump of about 0.5 in the probability of being enrolled in the *Juntos* program when the IFH score crosses the threshold. On the other hand, although the reduced form plots do not reveal obvious discontinuities in the probability of using an informal health care provider when ill at the IFH cutoff for *Juntos* eligibility, the implied treatment effect is found to be significant.

Table 2 presents instrumental variable estimates of τ —the effect of the *Juntos* program on the probability of using formal and informal care when ill—for the target and nontarget populations. The baseline results from Figure 1 correspond to column 4 with no controls. In panel a, the estimated effect is sizeable for the target group: the probability that an ill individual from the *Juntos* target group will seek medical attention at an informal provider is reduced by 0.35. Even though the CCT program does not explicitly require beneficiaries to seek formal medical care when ill, it does require them to attend regular medical checkups at the preestablished formal facility. Thus, the beneficiaries and their families become familiar enough with the facility itself and the staff running the facility to develop the habit of attending there, at least when it comes to child care, which thereby reduces the demand for informal health care. In panel b, the estimated effect is smaller (about 0.19) but remains significant for the nontarget group. Thus, the above conclusion regarding the decrease in the use of informal health care applies to all household members, not only beneficiaries.

Table 2 also shows results under various specifications, namely, different polynomial degrees d , bandwidths chosen optimally following Calonico et al. (2014), bandwidths chosen to be arbitrarily below and above the optimal value, and

¹⁰To illustrate, we use $d = 2$ and h chosen optimally using the criterion proposed by Calonico et al. (2014). The estimates of the reduced form effect were significant and robust to alternative specifications of d and h .

TABLE 2 Estimation of the effect of *Juntos* on the use of informal health care providers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(a) Target population (direct effects)								
No controls	−0.217*** (0.081)	−0.309*** (0.096)	−0.213*** (0.084)	−0.347*** (0.124)	−0.297*** (0.127)	−0.343*** (0.107)	−0.345*** (0.115)	−0.249*** (0.105)
With controls	−0.260*** (0.084)	−0.361*** (0.096)	−0.252*** (0.085)	−0.399*** (0.127)	−0.362*** (0.120)	−0.389*** (0.109)	−0.365*** (0.115)	−0.286*** (0.106)
Polynomial degree (<i>d</i>)	1	1	1	2	2	2	3	3
Bandwidth (<i>h</i>)	0.46	0.36	0.55	0.84	0.70	0.98	1.59	1.79
Observations (<i>n</i>)	3,210	2,768	3,291	4,404	3,947	4,671	5,360	5,411
(b) Nontarget population (indirect effects)								
No controls	−0.204*** (0.057)	−0.232*** (0.058)	−0.164*** (0.054)	−0.186*** (0.079)	−0.190* (0.104)	−0.235*** (0.065)	−0.227*** (0.071)	−0.174*** (0.065)
With controls	−0.232*** (0.060)	−0.264*** (0.061)	−0.185*** (0.056)	−0.234*** (0.078)	−0.243*** (0.103)	−0.250*** (0.066)	−0.259*** (0.071)	−0.237*** (0.063)
Polynomial degree (<i>d</i>)	1	1	1	2	2	2	3	3
Bandwidth (<i>h</i>)	0.42	0.38	0.46	0.69	0.54	1.03	1.59	1.98
Observations (<i>n</i>)	4,545	4,413	4,984	6,298	5,116	7,719	8,972	9,197

Source: ENAHOR 2014. Own elaboration. *Note.* Instrumental variable estimates of the treatment effect on the use of informal health care providers, τ in system (1), for various values of *d* and *h*. Robust standard errors are in parentheses. All regressions include a flexible polynomial of degree *d* with interaction terms with *T* (the eligibility indicator), as in Equation (2). *h* is the bandwidth, and *n* is the number of observations such as $-h \leq X \leq h$, where *X* is the IFH. In columns 1, 4, and 7, the bandwidth is chosen optimally (for the regressions with controls) following Calonico et al. (2014); in columns 2 and 5, the bandwidth is set deliberately below, whereas in columns 3, 6, and 8, it is set above the optimal value. The regressions with controls include individual characteristics such as female indicator, age and age squared, maximum education level achieved, an illiterate indicator, and indicators for the mother tongue (Spanish, Quechua, Aymara, and others). They also contain indicators of whether the town is considered an indigenous community, the distance (both in meters and in time) to the closest health center, and an indicator of whether the household is covered by public health insurance.

* Statistical significance at 10% confidence level.

** Statistical significance at 5% confidence level.

*** Statistical significance at 1% confidence level.

regressions with or without control variables. The estimates do not change substantially with the inclusion of control variables¹¹ and are remarkably robust to the values of *d* and *h*.¹²

4.2 | Heterogeneous effects

We now study potential heterogeneities in the estimated effects to understand the mechanisms behind our baseline results. We focus on the spillover effects of *Juntos* on nontarget members within the household. Table 3 shows the treatment effect τ under various specifications (i.e., different polynomial degrees *d* and bandwidths *h*) for several subsamples.

In panel a of Table 3, we classify the individuals according to land ownership, because ownership is a covariate of income and living conditions that the government does not explicitly use to compute the IFH. In the sample, there are 5,347 out of 9,625 individuals who belong to a household whose head reports is the owner of a plot and 4,278 individuals from landless households. The estimates show the importance of an income effect in the decision to replace informal services with formal ones, because the estimated effects are concentrated among those households that do not own any

¹¹Thus, the differences in the individual characteristics and other control variables cannot explain the discontinuity in the probability of benefiting from *Juntos* at the cutoff point. This is a condition for the satisfactory identification of the treatment effect of interest. In addition, following a standard empirical practice (see, inter alia, Amarante, Manacorda, Miguel, & Vigorito, 2016; Pinotti, 2017; Bergolo & Galván, 2018), we run the McCrary (2008) test and could not reject the hypothesis of continuity of the running variable around the cutoff point. This result indicates that the possible manipulation of the IFH is not able to explain the discontinuity in the probability of using *Juntos* either. This analysis is reported in Section A of the Supporting Information.

¹²In Section C of the Supporting Information, we show that our baseline results, and thus the interpretation we give to them, are robust to further sensitivity analyses. In particular, (a) the baseline results are conditional on illness, and we find similar effects when conditioning on medical attendance; (b) the results remain if we allow the supply-side indicators in the baseline results to be endogenous and thus exclude them from the list of control variables; and (c) the results do not change in a meaningful way when we restrict the sample to households that are covered by health insurance, which operates exclusively in formal facilities.

TABLE 3 Heterogenous effects of *Juntos* on the use of informal health care providers for nontarget groups

	Polynomial degree (<i>d</i>)						<i>t</i> ₂
	1	2	3	1	2	3	
(a)	Land ownership			No land ownership			<i>t</i> ₂
Effect (τ)	−0.052 (0.055)	−0.117 (0.085)	−0.089 (0.087)	−0.238*** (0.068)	−0.503*** (0.133)	−0.400*** (0.105)	3.593 [0.000]
Bandwidth (<i>h</i>)	0.63	0.94	1.74	0.63	0.84	1.89	
Observations (<i>n</i>)	3,509	4,340	5,082	2,651	3,134	4,020	
(b)	Children younger than 7 years			Children older than 7 years			<i>t</i> ₂
Effect (τ)	−0.247*** (0.084)	−0.317*** (0.096)	−0.298*** (0.097)	−0.116* (0.065)	−0.142** (0.081)	−0.120 (0.086)	1.968 [0.049]
Bandwidth (<i>h</i>)	0.49	1.07	1.86	0.58	0.91	1.87	
Observations (<i>n</i>)	3,069	4,570	5,195	2,108	3,152	3,939	
(c)	Other social programs			No other social programs			<i>t</i> ₂
Effect (τ)	−0.205*** (0.063)	−0.372*** (0.110)	−0.361*** (0.116)	−0.101** (0.050)	−0.182** (0.084)	−0.159** (0.074)	1.987 [0.047]
Bandwidth (<i>h</i>)	0.63	0.94	1.74	0.63	0.84	1.89	
Observations (<i>n</i>)	3,135	3,963	4,595	3,024	3,492	4,511	
(d)	Literate			Illiterate			<i>t</i> ₂
Effect (τ)	−0.159 (0.120)	−0.212 (0.180)	−0.194 (0.137)	−0.189*** (0.063)	−0.210*** (0.085)	−0.240*** (0.073)	0.013 [0.989]
Bandwidth (<i>h</i>)	0.49	0.60	1.79	0.48	0.63	1.82	
Observations (<i>n</i>)	1,618	1,711	3,205	3,385	4,105	5,903	
(e)	Male			Female			<i>t</i> ₂
Effect (τ)	−0.139 (0.092)	−0.200 (0.126)	−0.262** (0.131)	−0.197*** (0.070)	−0.303*** (0.092)	−0.257*** (0.089)	0.953 [0.341]
Bandwidth (<i>h</i>)	0.52	0.67	1.43	0.51	0.76	1.62	
Observations (<i>n</i>)	2,306	2,806	3,933	2,777	3,642	4,938	

Source: ENAHOR 2014. Own elaboration. *Note.* Instrumental variable estimates of the treatment effect on the use of informal health care providers, τ in system (1), for various values of *d* and *h* and for subsamples. Robust standard errors are in parentheses. All regressions include a flexible polynomial of degree *d* with interaction terms with *T* (the eligibility indicator) as in Equation (2), and the control variables are listed in the notes of Table 2. *h* is the bandwidth, chosen as in Calonico et al. (2014), and *n* is the number of observations such as $-h \leq X \leq h$, where *X* is the IFH. The column *t*₂ is the absolute *t* statistic of the null $H_0: \tau_1 = \tau_2$, where τ_i is the effect in subsample *i*, for *d* = 2. The *p* values are shown in square braces, and the results for different *d* (available upon request) are similar.

*Statistical significance at 10% confidence level.

**Statistical significance at 5% confidence level.

***Statistical significance at 1% confidence level.

land, with the difference between effects in the two subsamples being statistically significant (see the *t*₂ column). Budget and time constraints are still major barriers to the use of institutional health services, and the cash transfer can help alleviate such constraints. In particular, the cash transfer can be enough to cover the opportunity cost of members of destitute households joining the queue for formal health care.

However, as can be seen from the rest of the table, budget considerations are not the only mechanism driving the results. We show compelling evidence that the availability of better information about institutional services and especially the greater confidence generated by frequent interaction with formal health facilities are potential mechanisms driving our main effects.

First, even though health conditionalities of *Juntos* are mandatory only for children under 3 years of age, the government (through MIDIS or MINSA) often launches nutritional and health campaigns aimed at preschoolers and the youngest schoolers (see Escobal & Benites, 2012). Thus, households with young children are more likely to become familiar with formal health care faster. In panel b, we show the number of individuals that belong to households with at least one child below 7 years old (5,475) and those that do not (4,150). The panel also shows that the effects of the first group are stronger as a formal *t* test cannot reject the null hypothesis of equal effects. These findings indicate that frequent interaction with formal health facilities their staffs can lead to increased familiarity with these services. In time, this familiarity leads to attending formal health facilities once the individual is ill.

By the same token, panel c shows the individuals from households who are beneficiaries of other social programs and therefore have interacted previously with government representatives (4,818) and those who are not (4,806). The estimated effects are significant regardless of whether the ill individual belongs to a household that is targeted by another social program or not. The magnitude, however, is substantially larger if the household had previously received the assistance from other governmental social programs, and statistically different. This result is evidence that greater interaction with public services generates greater confidence among rural inhabitants, which in turn leads to a reduction in the use of informal health services.

Even though suggestive, the evidence shown in the remaining cases is less clear-cut. In panel d, the sample is divided between people able to read (3,362) and illiterates (6,263). We use these classifications as proxies for informed and uninformed people. The estimated effects are significant for the illiterates, who presumably are those most affected by the information shock associated with the CCT program. Nonetheless, the effects for the literates are imprecisely estimated, and, because of this, we cannot reject the hypothesis of equal effects in both groups (i.e., the t statistic for this difference is rather small).

Similarly, panel e divides the population by gender, with women (5,292) being most likely to interact with the formal health system and to obtain more first-hand information about the quality of their services. This is so because most of the time women take care of children, or are themselves beneficiaries. Thus, the effects on women should be stronger than on men, which is precisely what we obtain in the estimation. The differences, however, are not statistically significant as the effects for men are estimated quite imprecisely.

5 | CLOSING REMARKS

We have provided robust evidence on a negative effect of a CCT program on the demand for informal health care in rural Peru. This effect is in part an income effect, as the extra cash received by the household covers the cost of getting medical attention in institutional health facilities, such as the actual transport costs and the opportunity cost of queueing. Nonetheless, it also is due to the acquisition of better information and, possibly, a different attitude toward modern medicine. CCT requires medical checkups for children and pregnant women in formal health care facilities, and so, the regular and frequent interactions of the members of a treated household will help develop the habit of consuming formal health care instead of an informal, low-quality alternative.

These results provide new insights for policymakers on how CCT programs induce behavioral responses in every member of treated households. By changing individual's behaviors, these interventions are able to effectively reshape health systems in poor countries. This evidence also indicates that some of the main effects of CCTs might not be necessarily due to the program's medical checkups but actually through the change in attitude toward formal health services that these programs encourage. Estimating the effect of these programs without considering these aspects would underestimate its total impact.

Due to data limitations, we cannot assess whether this reduction in the demand for informal health care due to CCTs is associated to an actual improvement in health. Similarly, we are not able to examine if the preference for formal services is focused on specific types of health services, which would help us to understand more about this shift in preferences. Finally, given our estimates on spillover effects within the household, it would be particularly relevant to estimate whether CCTs are also able to generate positive indirect effects among households. Examining these issues is beyond the scope of this paper but warrants future research.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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